

## 3.2

**a**

$$p(k) = ck$$

We know that a probability mass function is the probability that  $X = k$  and that the  $\sum_{k=1}^6 p(k) = 1$

Using this:

$$\sum_{k=1}^6 p(k) = 1$$

$$\sum_{k=1}^6 ck = 1$$

$$\rightarrow c + 2c + 3c + 4c + 5c + 6c = 1$$

$$\rightarrow 21c = 1$$

$$c = \frac{1}{21}$$

**b**

$$O = X \in \{1, 3, 5\}$$

$$P(O) = p(1) + p(3) + p(5)$$

$$P(O) = \frac{1}{21} + \frac{3}{21} + \frac{5}{21}$$

$$P(O) = \frac{9}{21} = \frac{3}{7}$$

## 3.3

**a**

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^0 f(x)dx + \int_0^{\infty} f(x)dx$$

$$\rightarrow 0 + \int_0^{\infty} 3e^{-3x}dx$$

$$\rightarrow -e^{-3(\infty)} - (-e^{-3(0)})$$

$$\rightarrow 0 + e^0$$

$$\rightarrow 1$$

Therefore this is a valid probability density function because  $\int f(x)dx = 1$

**b**

$$\begin{aligned}P(-1 < X < 1) &= \int_{-1}^1 f(x)dx = \int_{-1}^0 f(x)dx + \int_0^1 f(x)dx \\&\rightarrow 0 + \int_0^1 3e^{-3x}dx \\&\rightarrow -e^{-3(1)} - (-e^{-3(0)}) \\&\rightarrow -e^{-3} + e^0 \\&\rightarrow P(-1 < X < 1) = 1 - e^{-3} = 0.95021\end{aligned}$$

**c**

$$\begin{aligned}P(X < 5) &= \int_{-\infty}^5 f(x)dx = \int_{-\infty}^0 f(x)dx + \int_0^5 f(x)dx \\&\rightarrow 0 + \int_0^5 3e^{-3x}dx \\&\rightarrow -e^{-3(5)} - (-e^{-3(0)}) \\&\rightarrow -e^{-15} + e^0 \\&\rightarrow P(X < 5) = 1 - e^{-15} = 0.999999694\end{aligned}$$

**d**

$$\begin{aligned}P(2 < X < 4|X < 5) &= \frac{P(2 < X < 4 \cap X < 5)}{P(X < 5)} \\P(2 < X < 4 \cap X < 5) &= P(2 < X < 4) \\P(2 < X < 4|X < 5) &= \frac{P(2 < X < 4)}{P(X < 5)} \\P(2 < X < 4) &= \int_2^4 f(x)dx \\&\rightarrow \int_2^4 3e^{-3x}dx \\&\rightarrow -e^{-3(4)} - (-e^{-3(2)}) \\&\rightarrow -e^{-12} + e^{-6} \\&\rightarrow P(2 < X < 4) = e^{-6} - e^{-12} = 0.0024726 \\&\text{Take } P(X < 5) \text{ from part c}\end{aligned}$$

$$P(2 < X < 4 | X < 5) = \frac{e^{-6} - e^{-12}}{1 - e^{-5}}$$

$$P(2 < X < 4 | X < 5) = \frac{0.0024726}{0.999999694}$$

$$P(2 < X < 4 | X < 5) = 0.0024893812$$

## 3.7

**a**

$$[\sqrt{2}, \sqrt{3}]$$

**b**

$$P(X = 1.6) = 0 \text{ because } \lim_{a \rightarrow 0} F(1.6) - F(1.6 - a) = 0$$

**c**

$$P(1 < X < \frac{3}{2}) = F(\frac{3}{2}) - F(1) = \frac{3}{2}^2 - 2 - 0 = \frac{1}{4}$$

**d**

$$f(x) = \begin{cases} 0 & \text{if } x < \sqrt{2} \\ 2x & \text{if } \sqrt{2} \leq x < \sqrt{3} \\ 0 & \text{if } \sqrt{3} \leq x \end{cases}$$

## 3.9

**a**

$$\begin{aligned} \mu &= \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^0 xf(x)dx + \int_0^{\infty} xf(x)dx \\ &\rightarrow 0 + \int_0^{\infty} 3xe^{-3x}dx \\ &\rightarrow -\frac{(3(\infty)+1)e^{-3(\infty)}}{3} - \left(-\frac{(3(0)+1)e^0}{3}\right) \\ &\rightarrow 0 + \frac{(1)*1}{3} \\ \mu &= \frac{1}{3} \end{aligned}$$

### 3.25(b)

$h(x)$  is a pdf if  $\int_{-b}^b h(x)dx = 1$

$$\int_{-b}^b \cos(x)dx = \sin(b) - \sin(-b) = 1$$

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}, \sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$$

$$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}, \sin\left(-\frac{5\pi}{6}\right) = -\frac{1}{2}$$

$$\therefore \forall n \in \mathbf{Z} (b = \frac{\pi}{6} + 2\pi n) \vee (b = \frac{5\pi}{6} + 2\pi n)$$

### 3.32

**a**

$$P(X > 10) = \int_{10}^{\infty} f_X(x)dx$$

$$\int_{10}^{\infty} \frac{1}{2}x^{-\frac{3}{2}}dx$$

$$-(\infty)^{-\frac{1}{2}} - (-(10)^{-\frac{1}{2}})$$

$$0 + \frac{1}{\sqrt{10}}$$

$$\rightarrow P(X > 10) = \frac{1}{\sqrt{10}}$$

**b**

$$F_X(b) = P(X < b) = 1 - P(X > b)$$

$$\rightarrow 1 - \int_b^{\infty} f_X(x)dx$$

$$\rightarrow 1 - (-(\infty)^{-\frac{1}{2}} - (-(b)^{-\frac{1}{2}}))$$

$$\rightarrow 1 - (0 + \frac{1}{\sqrt{b}})$$

$$\rightarrow 1 - \frac{1}{\sqrt{b}}$$

$$F_X(x) = \begin{cases} 1 - \frac{1}{\sqrt{x}} & \text{if } 1 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

**c**

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f(x) dx \\ &= 0 + \int_1^{\infty} \frac{1}{2} x^{-\frac{1}{2}} dx \\ &= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{2} x^{-\frac{1}{2}} dx \\ &= \lim_{b \rightarrow \infty} \sqrt{b} - \sqrt{1} \\ &= \infty \\ \therefore E[X] &= \infty \end{aligned}$$